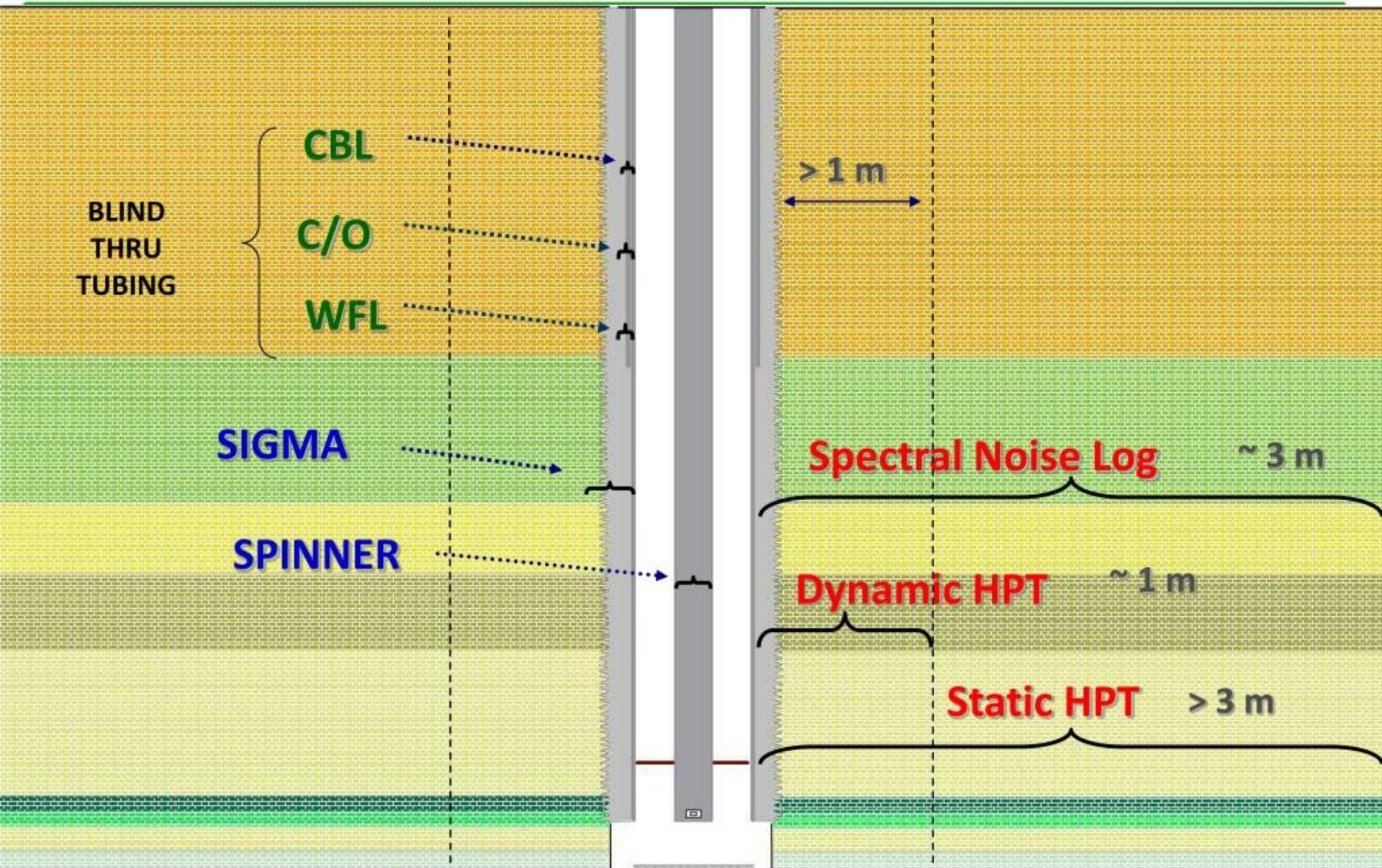
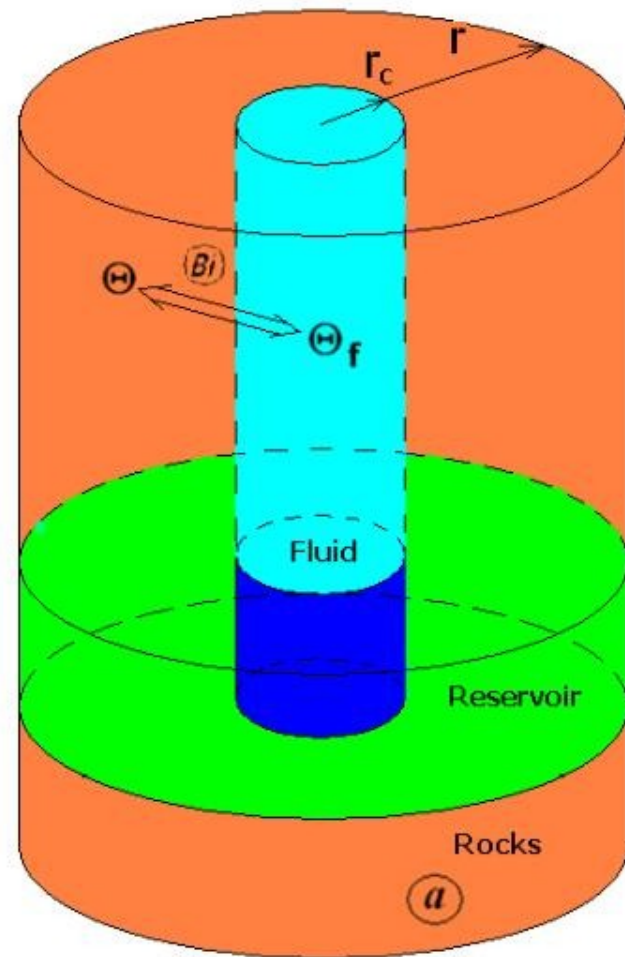
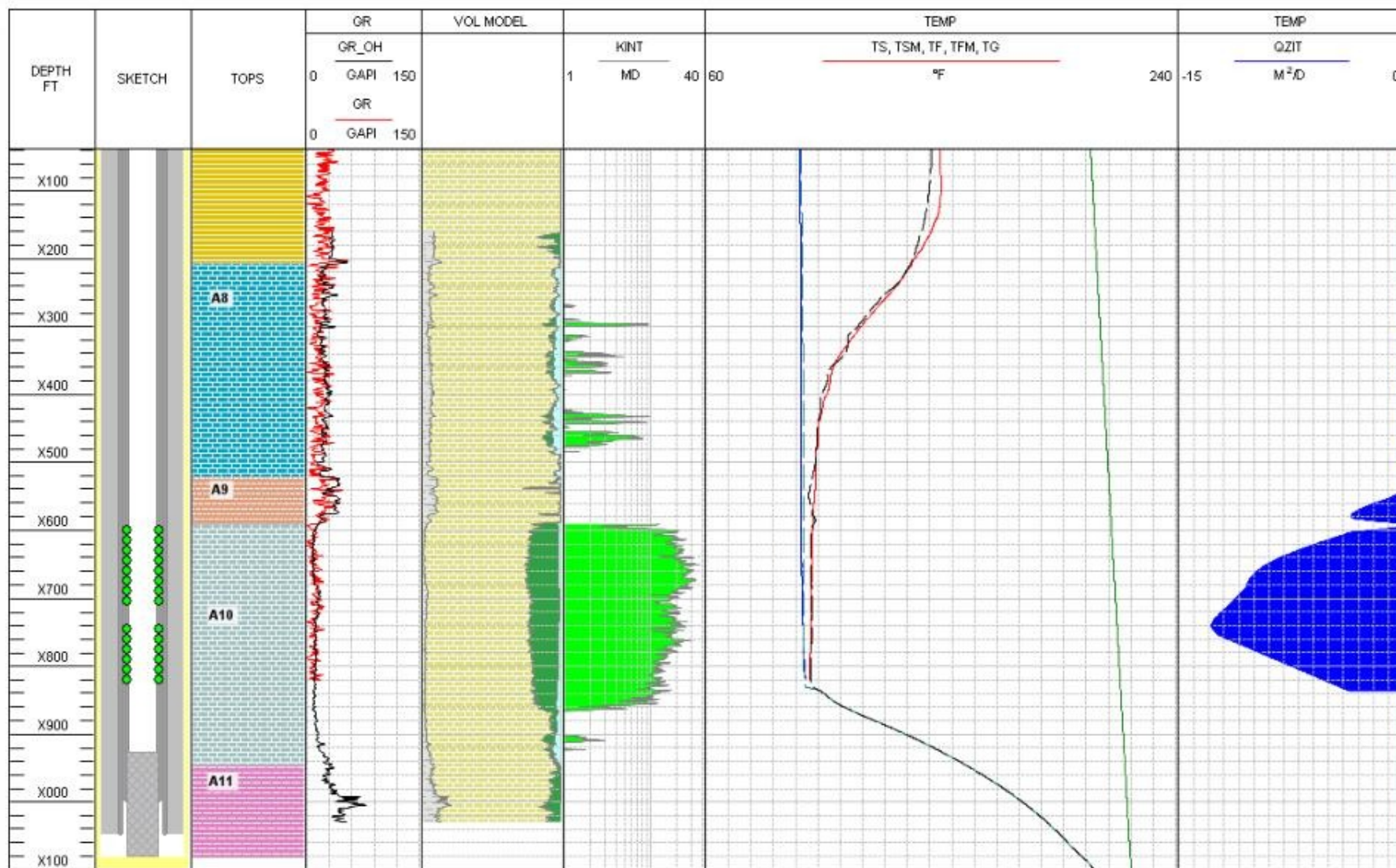


TEMPERATURE AND NOISE LOGGING IN RESERVOIR FLOW EVALUATION

Take a look behind the casing ...

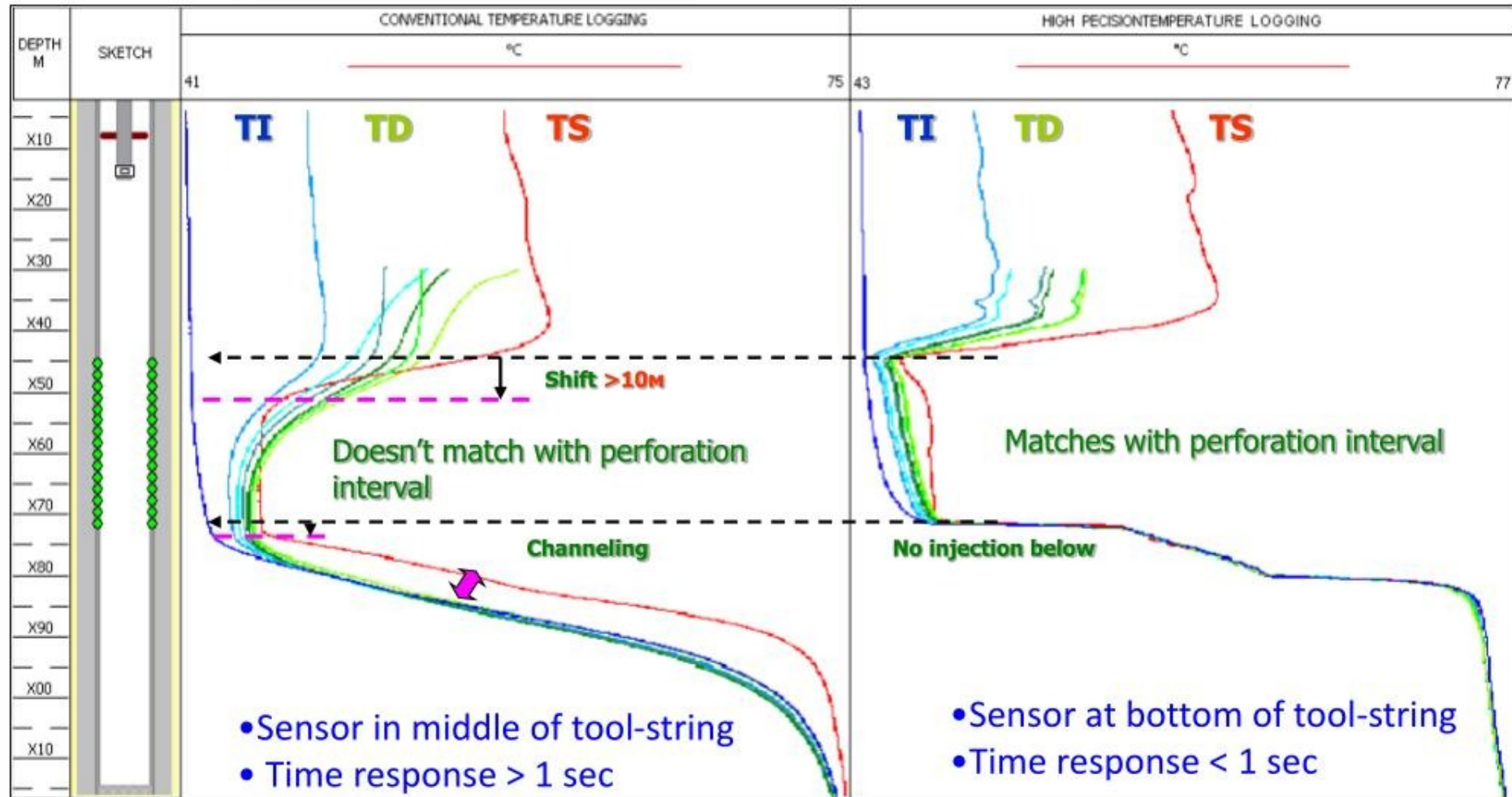


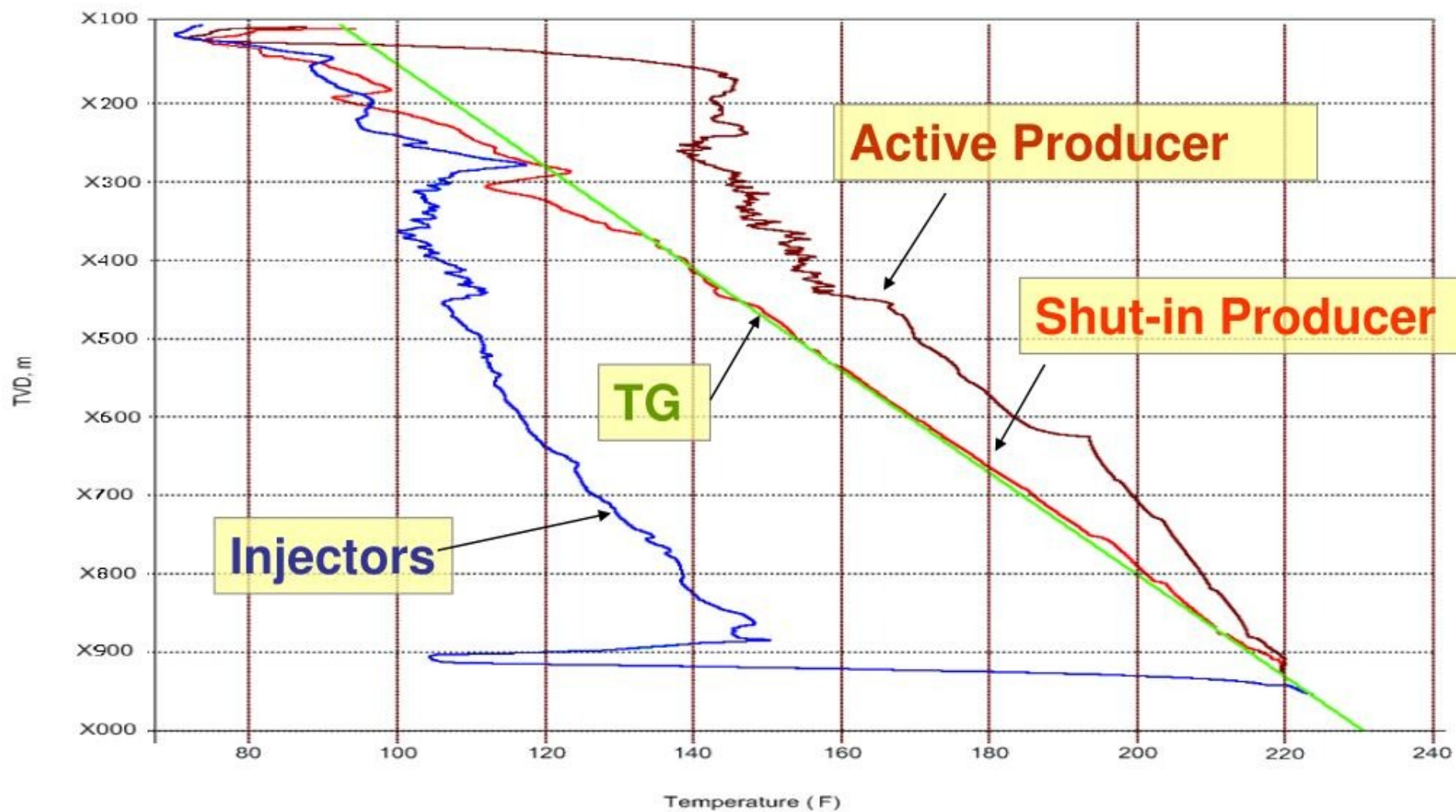


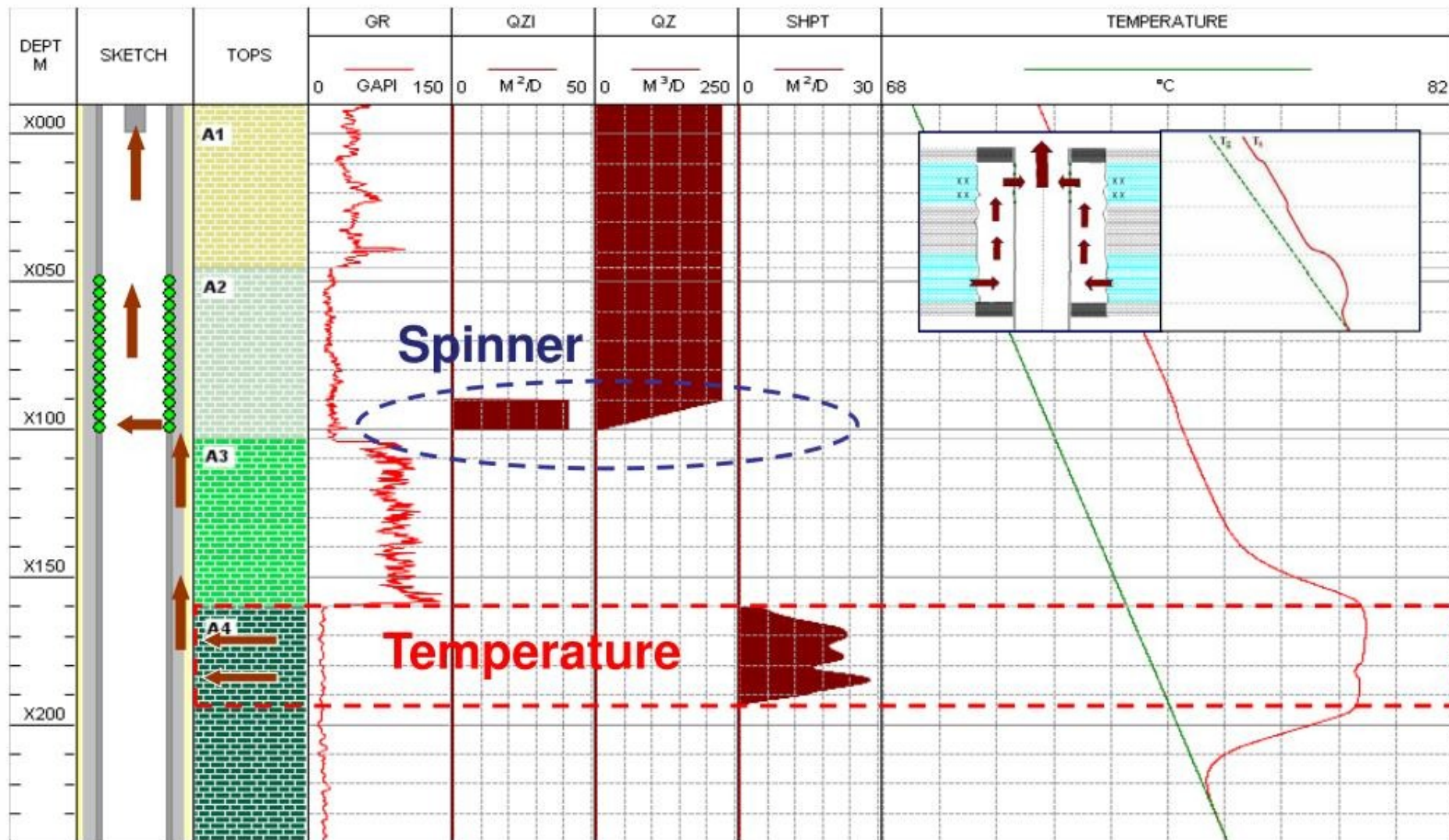


Conventional temperature

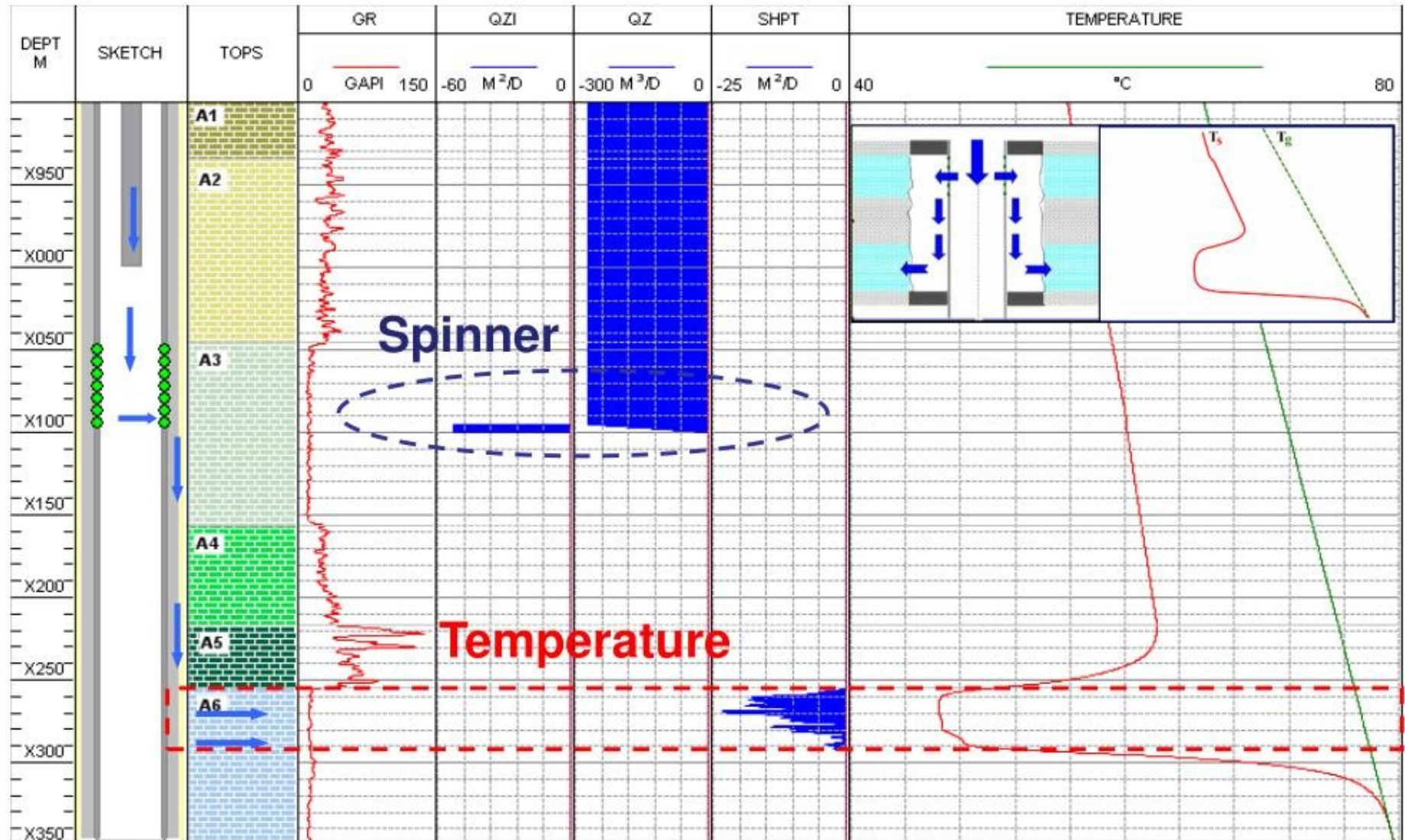
High precision temperature

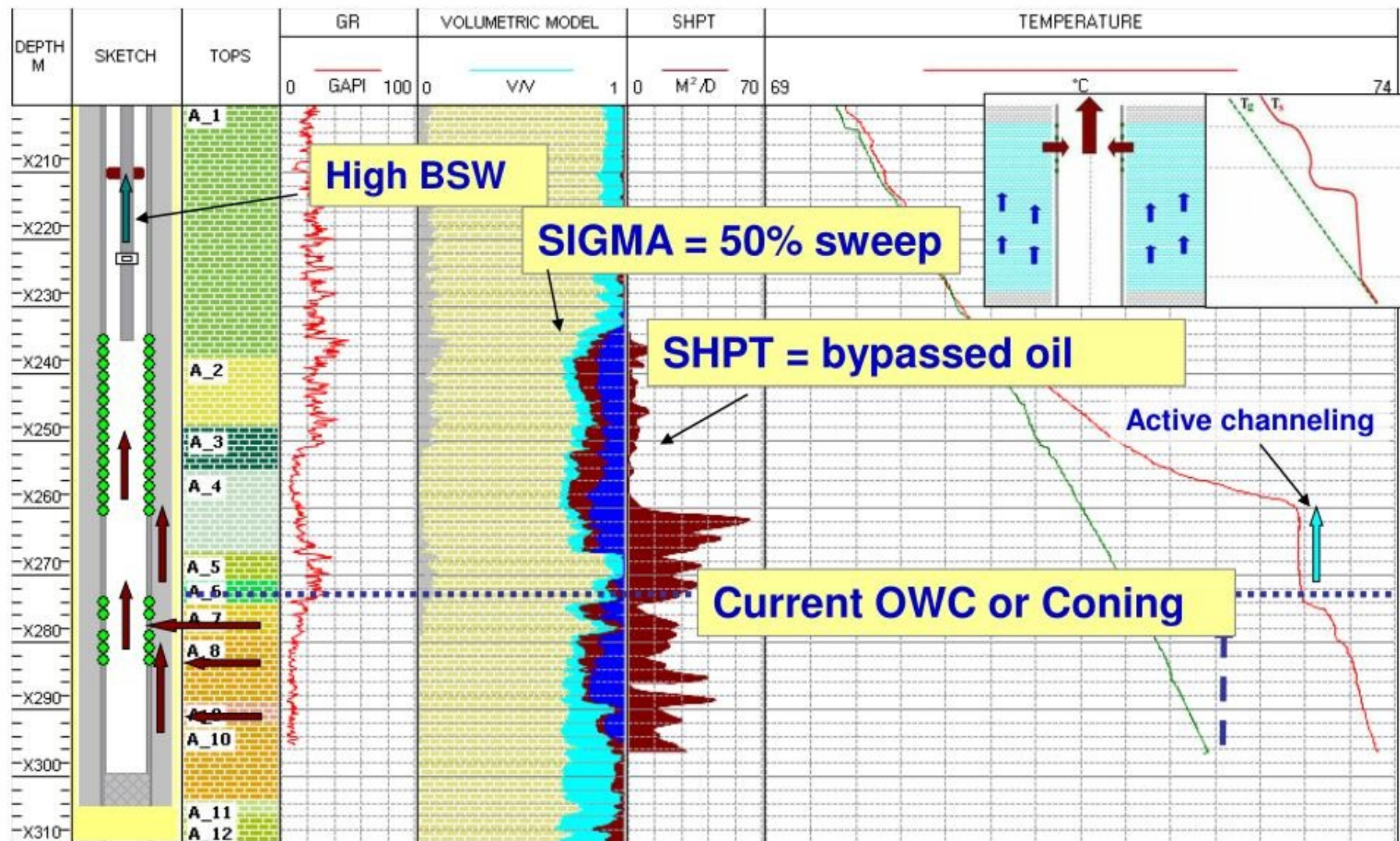




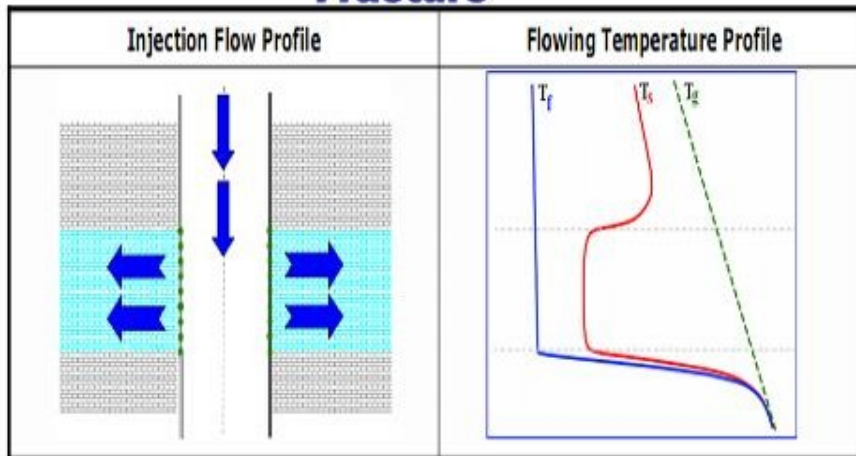


Injector. SHPT vs PLT

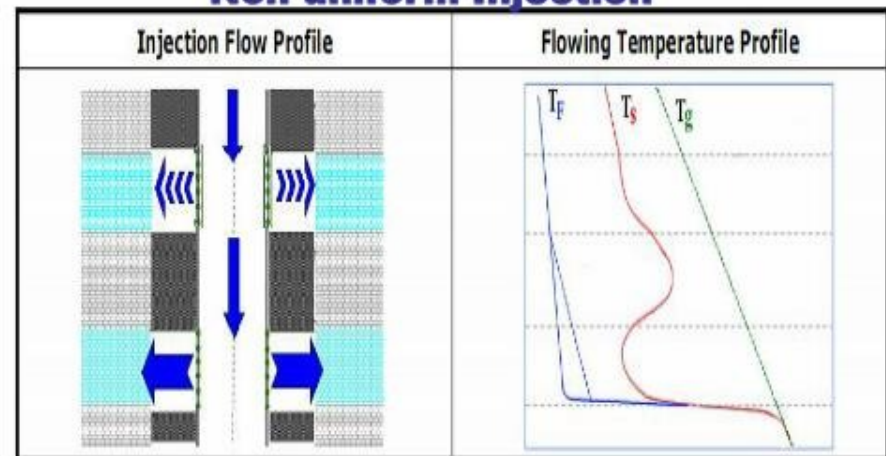




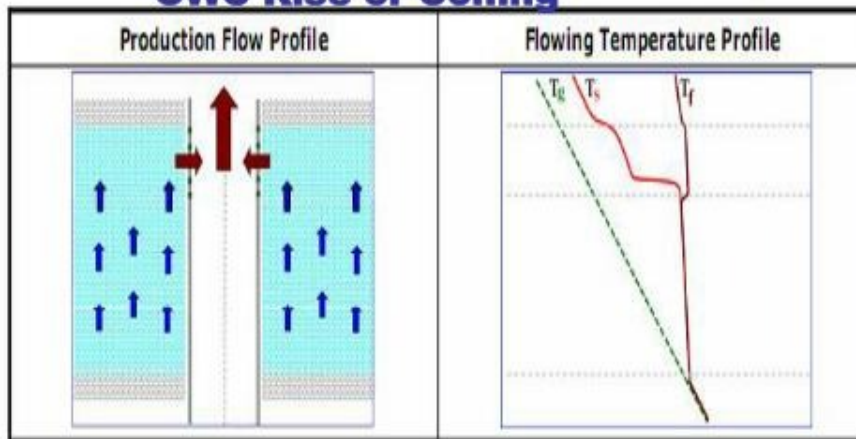
Fracture



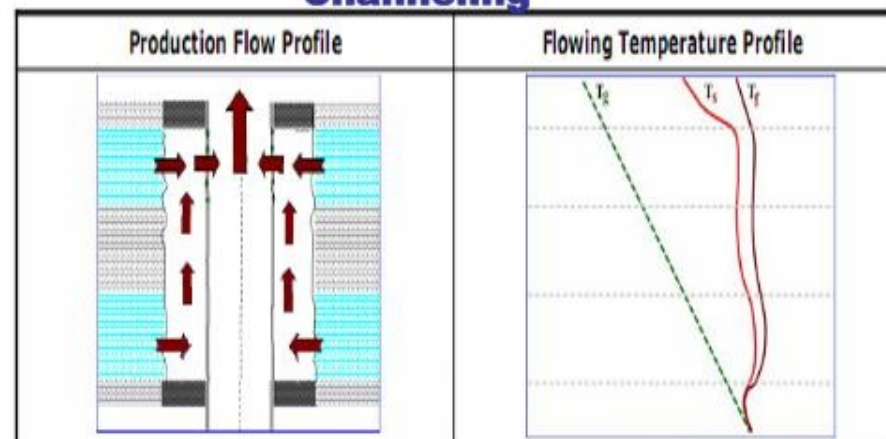
Non uniform Injection



OWC Rise or Coning



Channeling



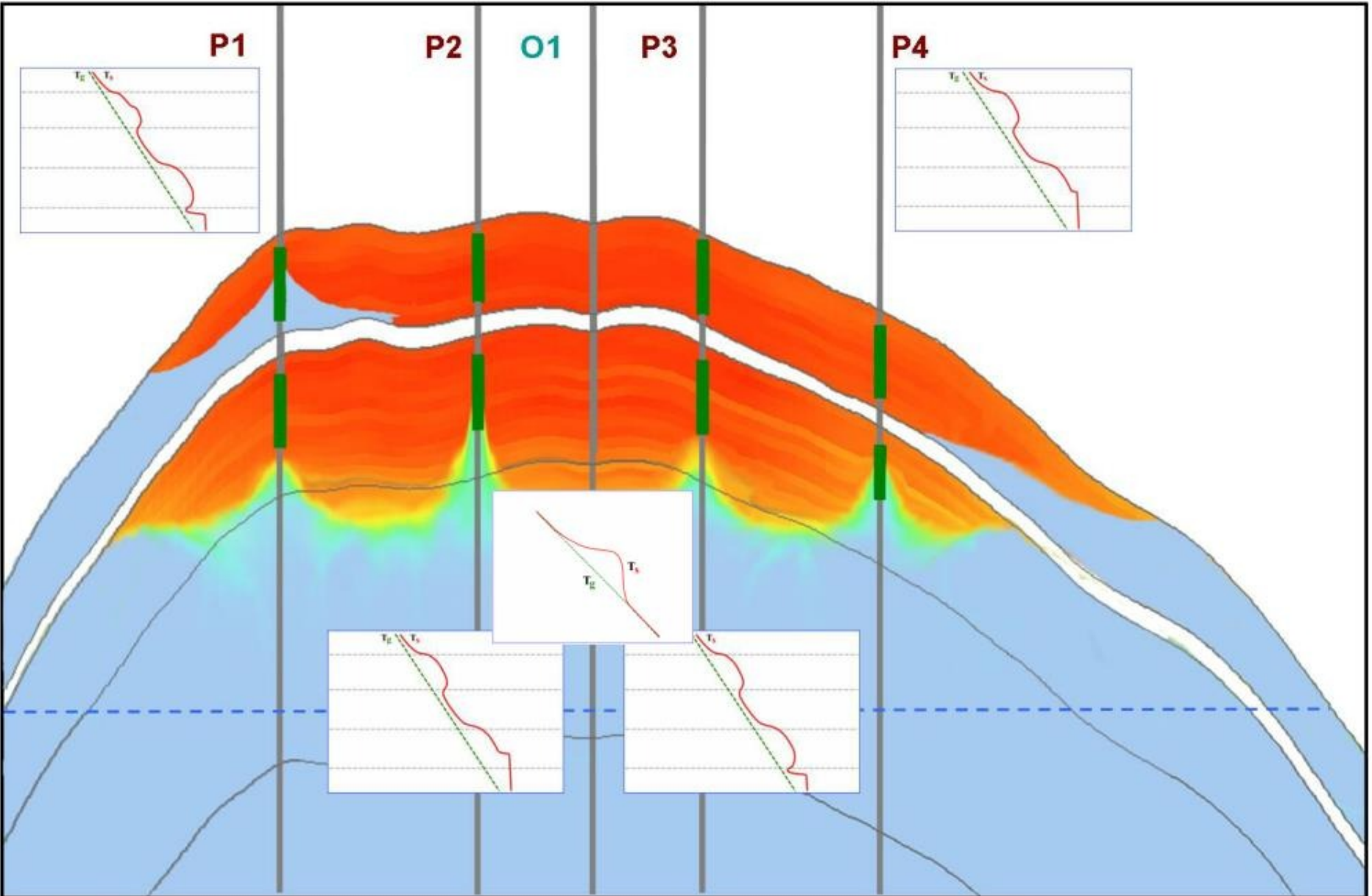
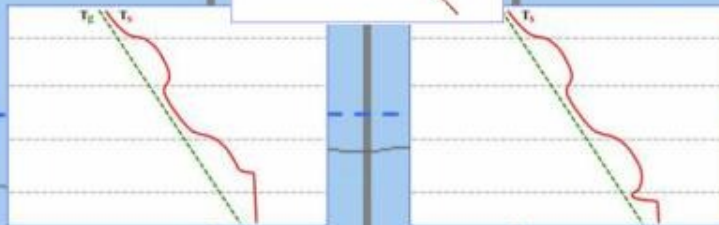
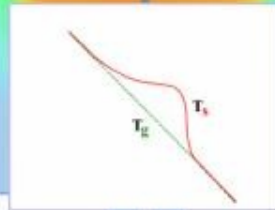
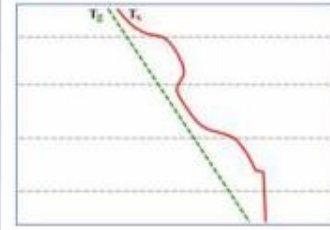
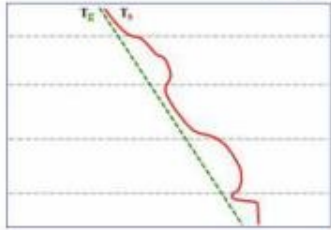
P1

P2

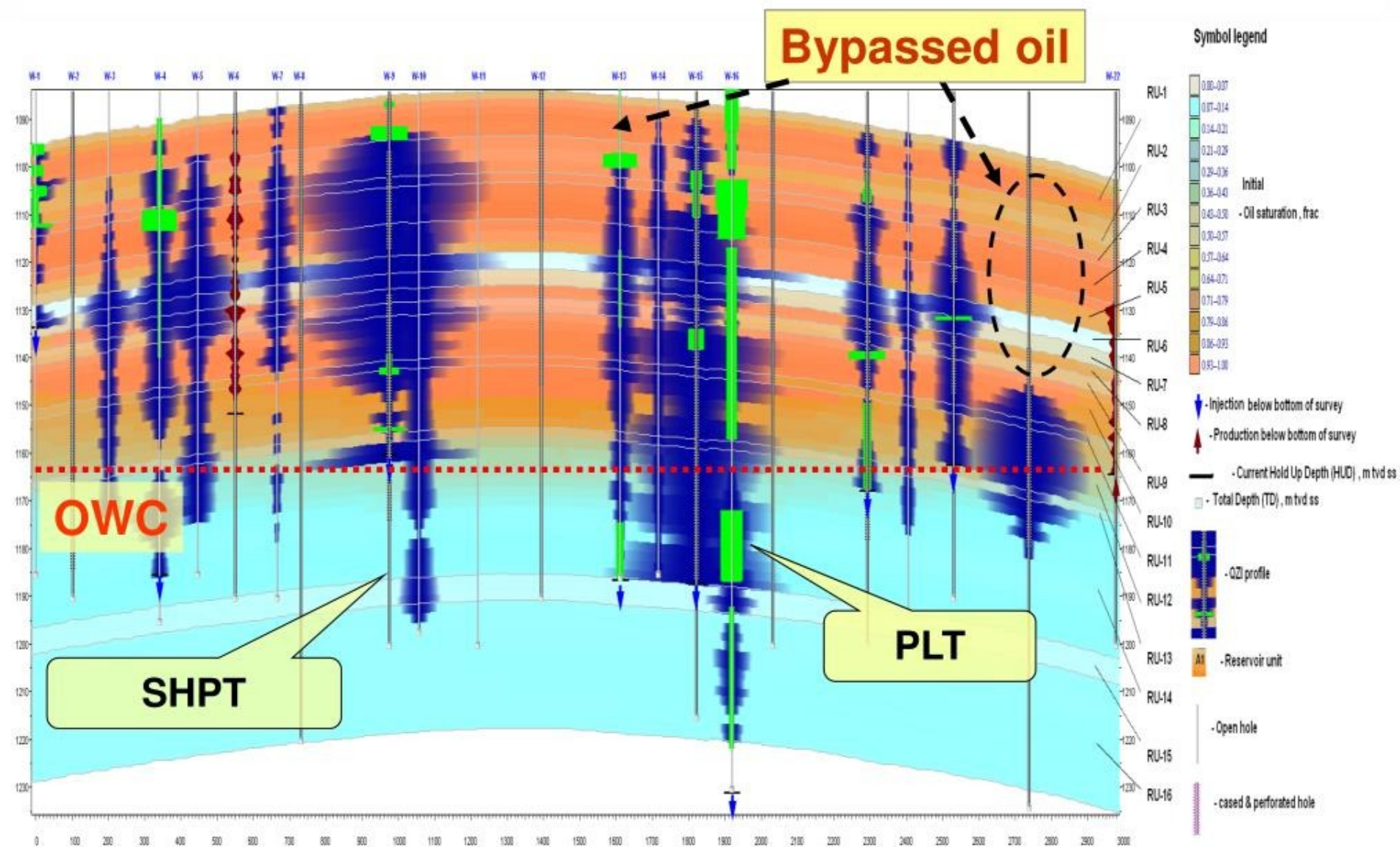
O1

P3

P4

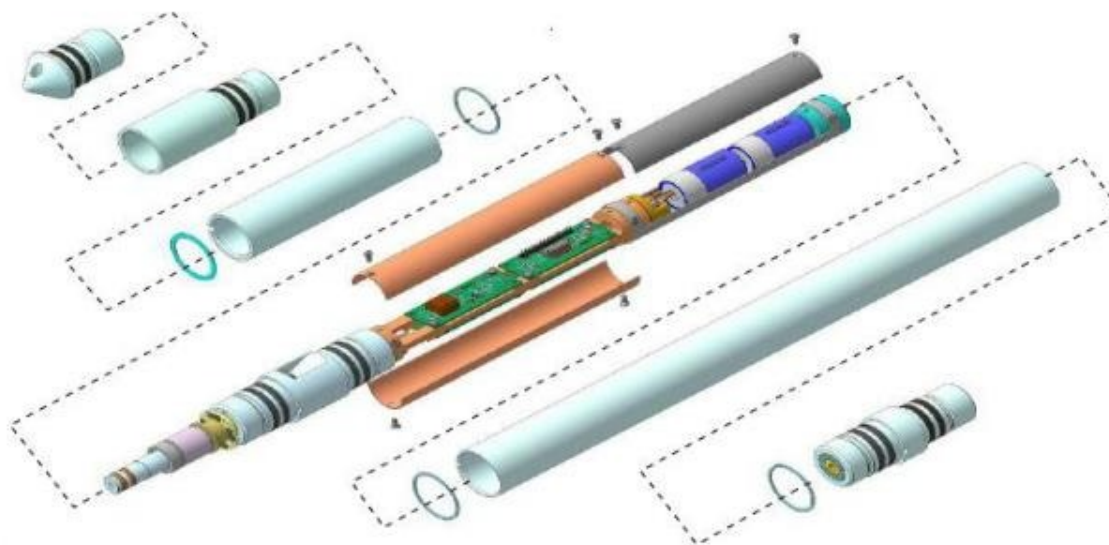


SHPT vs PLT 3D simulation



Parameter	Value
Operating frequency range, Hz	234.375 – 30,000
Frequency resolution, Hz	234.375
Number of frequency channels	512
Noise measurement range, dB	30–130
Maximum ambient operating pressure, MPa	60
Operating temperature range, degC	-20...+125

- **Reading Mode – Memory**
- **Temperature range – 125°C (253°F)**
- **Pressure Range – 60 MPa**



Noise Spectrum is defined by
console/reservoir structure

1 kHz

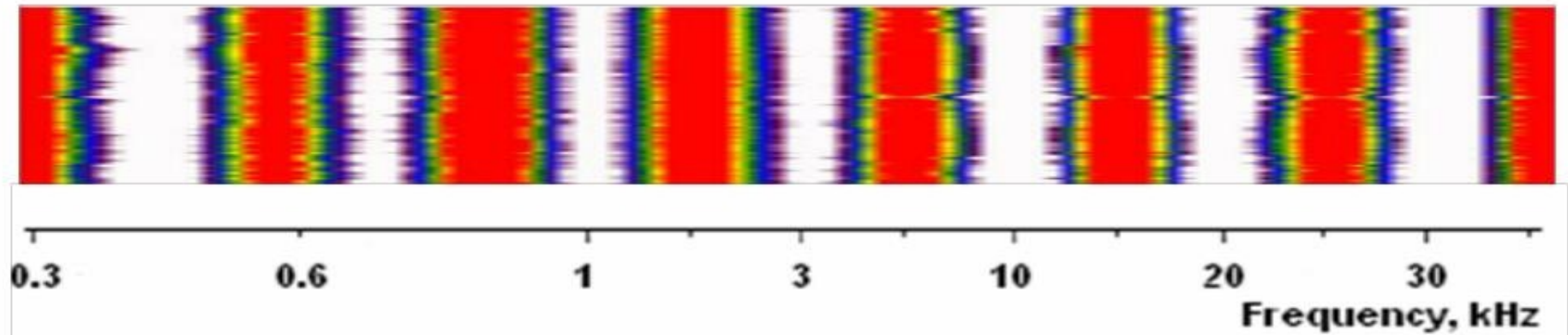


$$f = \frac{R}{l^2} \sqrt{\frac{E}{\rho}}$$

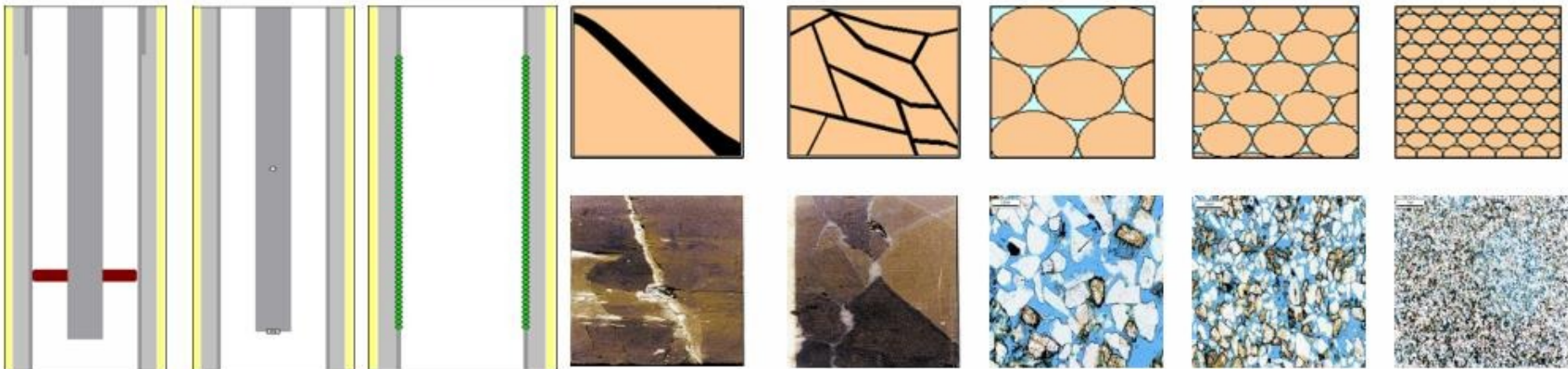
R	[m]	Diameter of console
l	[m]	Length of console
E	[Pa]	Young's modulus of console's material
ρ	[kg/m ³]	Density of console's material

Large consoles → Low frequencies

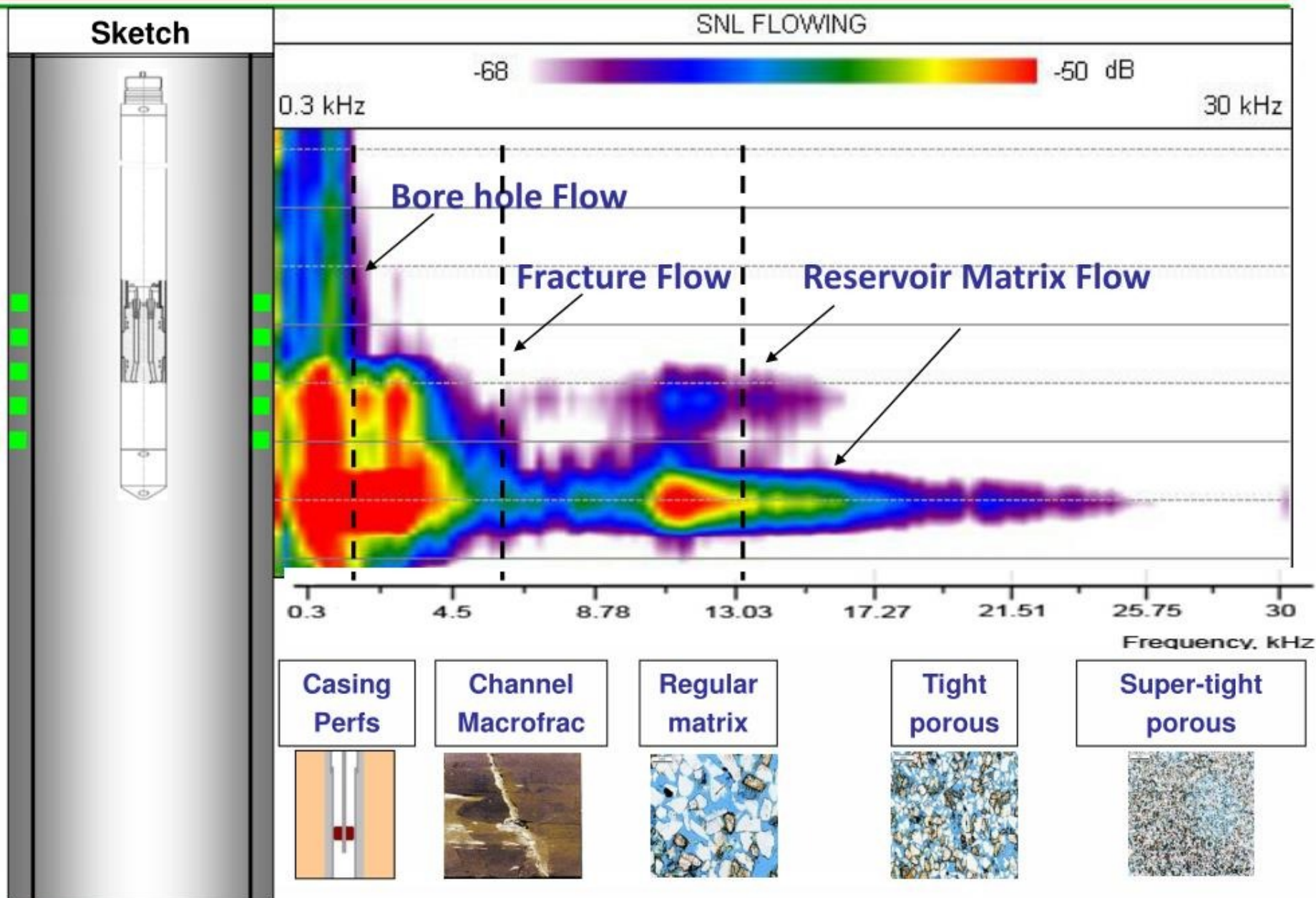
Small consoles → High frequencies

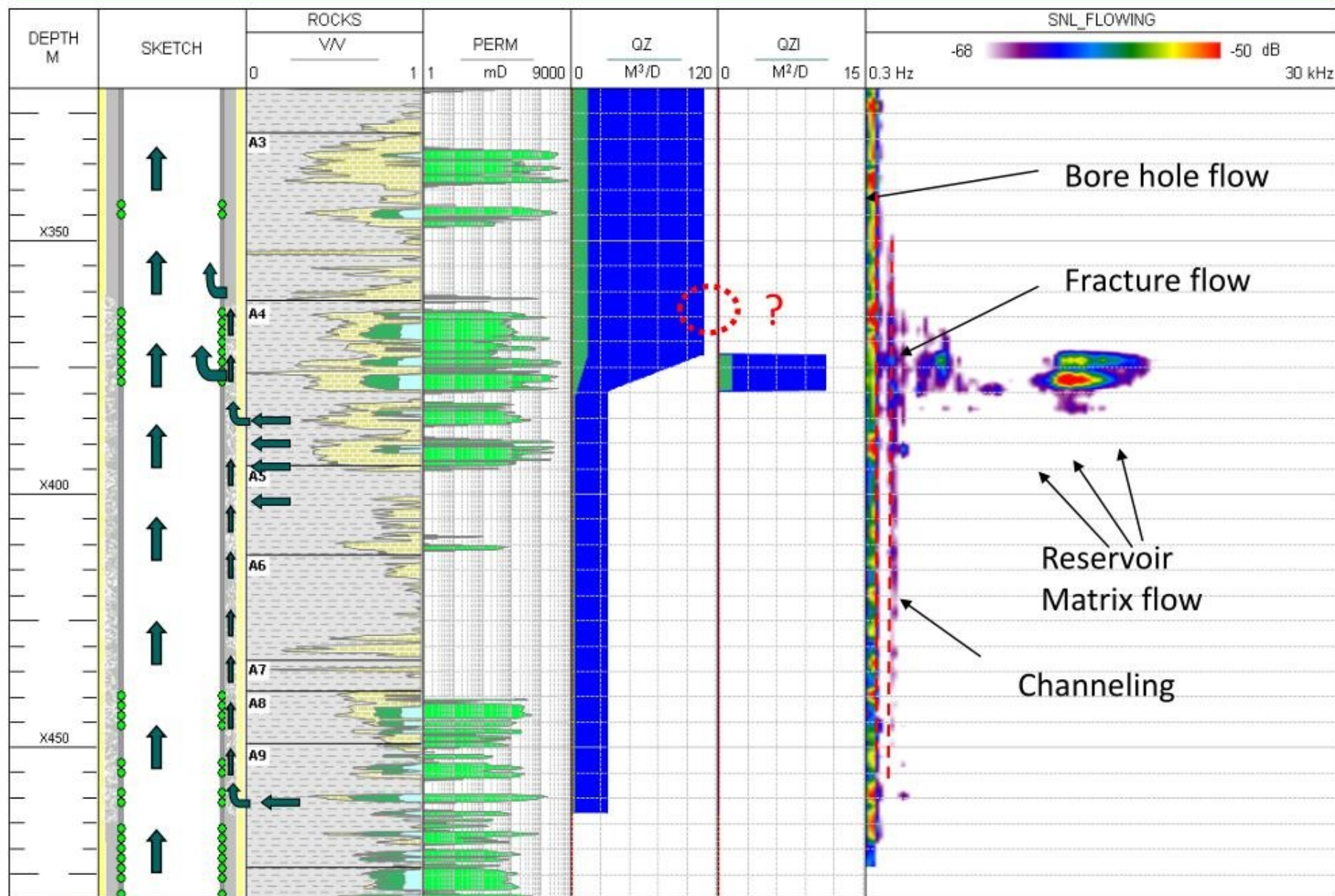


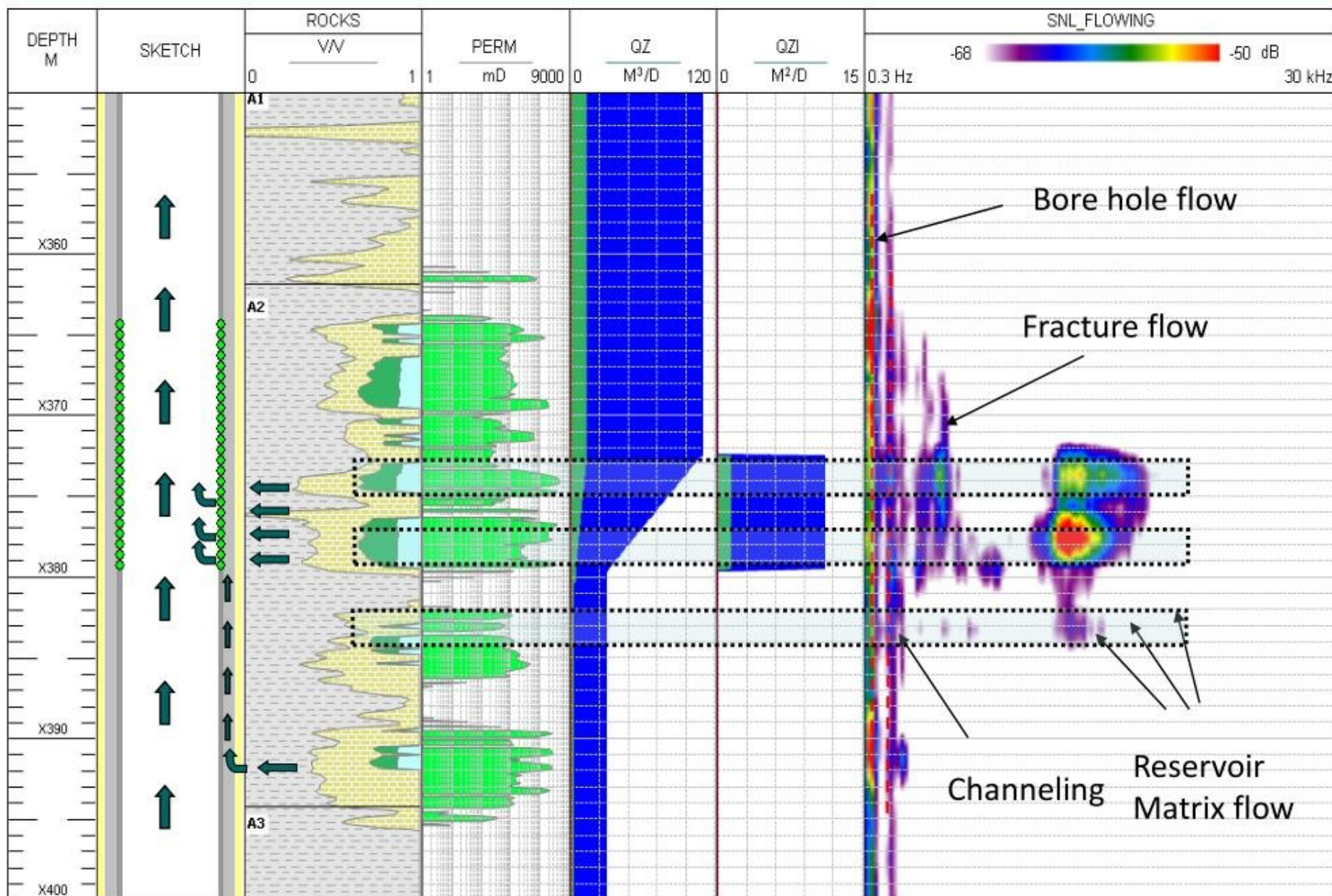
Casing Tubing	SSD X-NIPPEL Mandrels	Perfs Orifice Leaks	Channels Faults	Fractures	Regular matrix	Tight matrix	Super-tight Matrix
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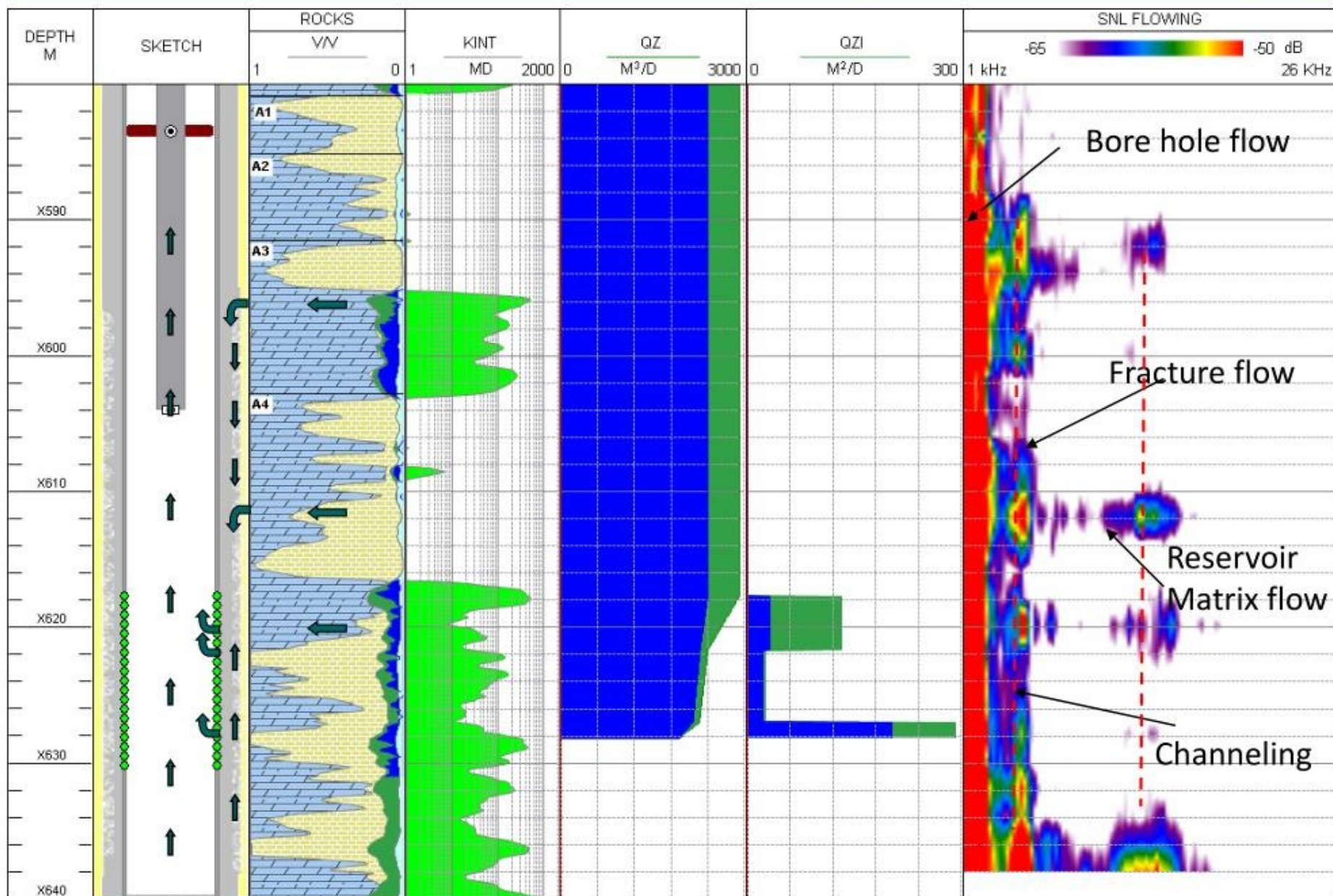


How to read SNL panel...

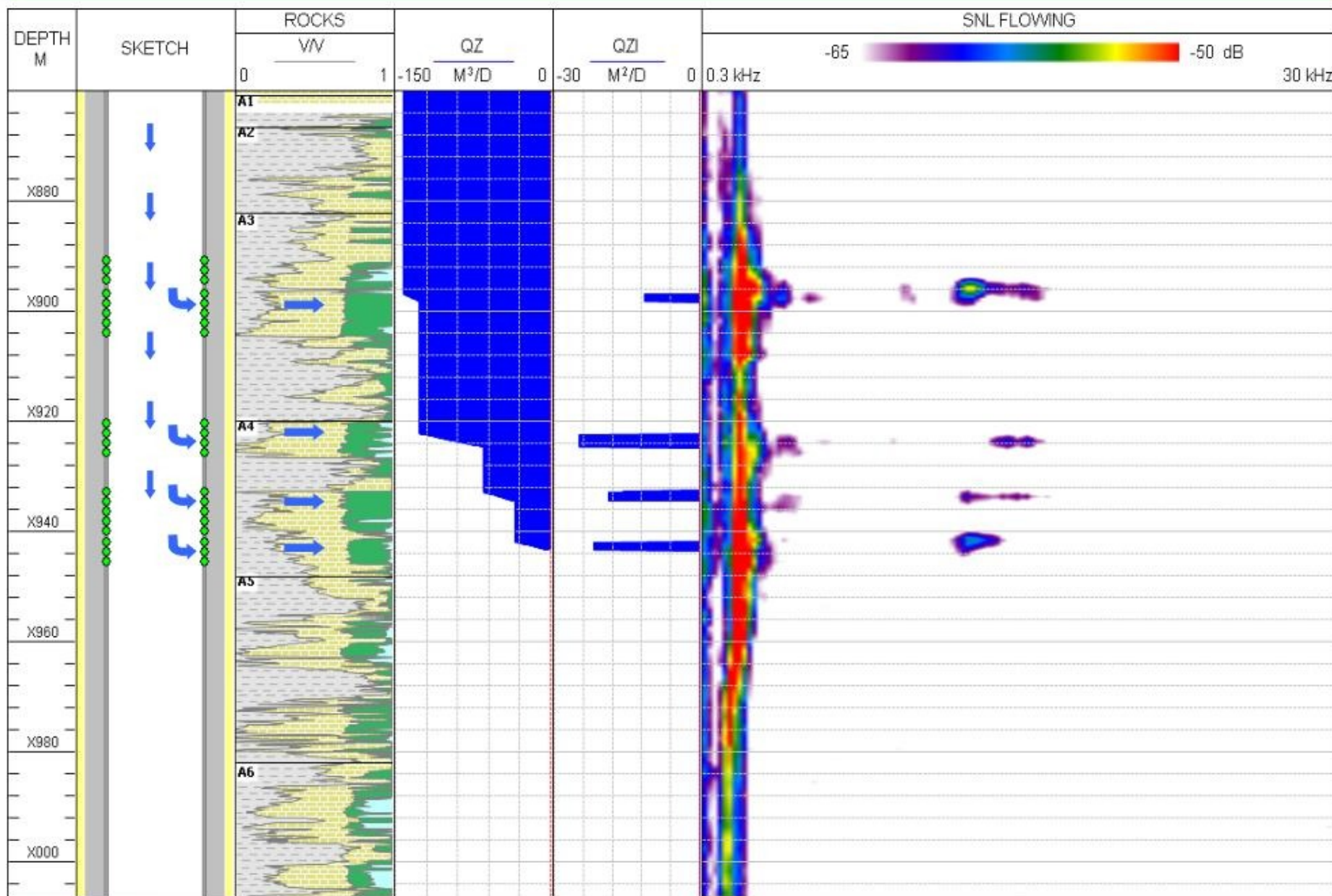




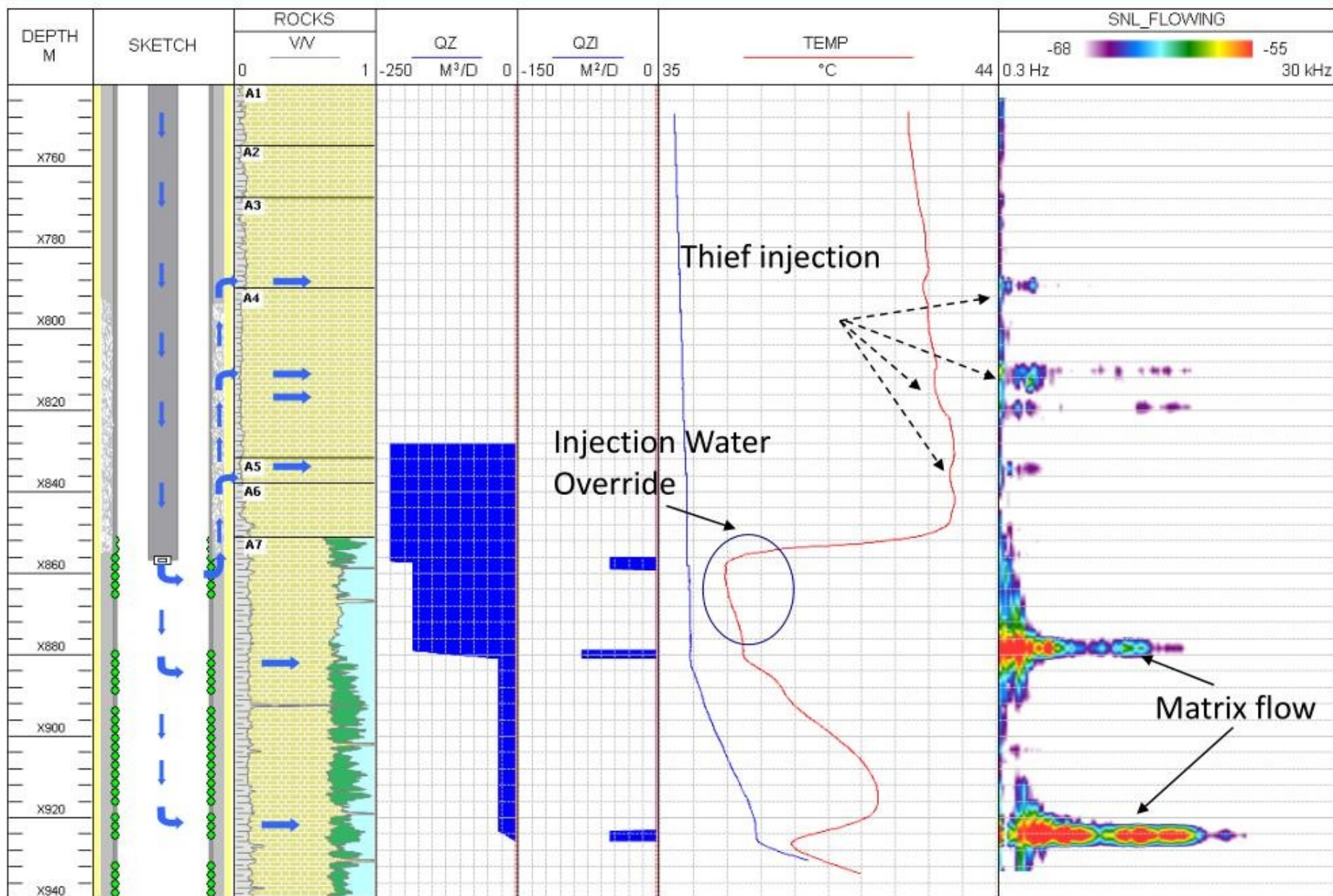


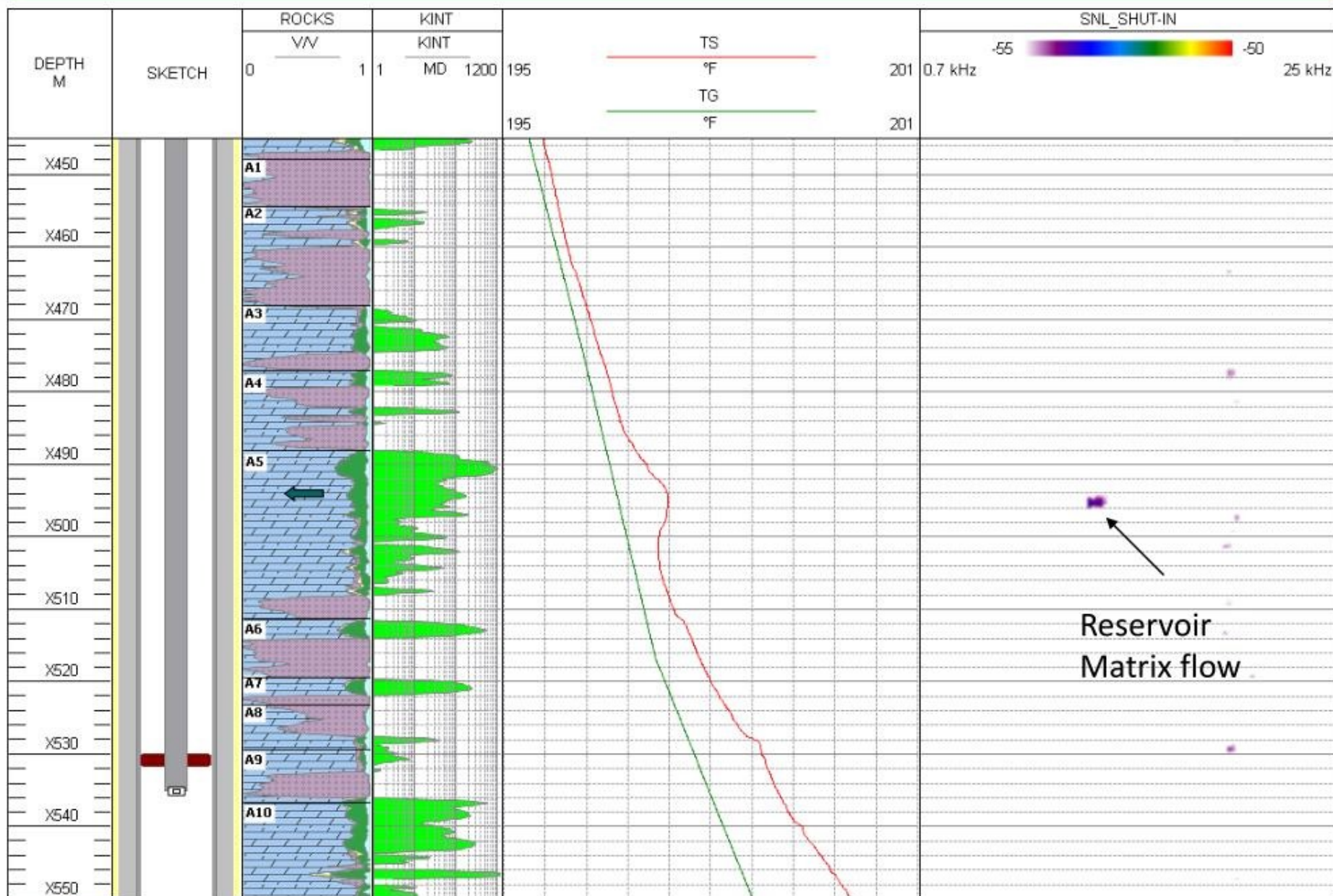


Injector. SNL vs PLT

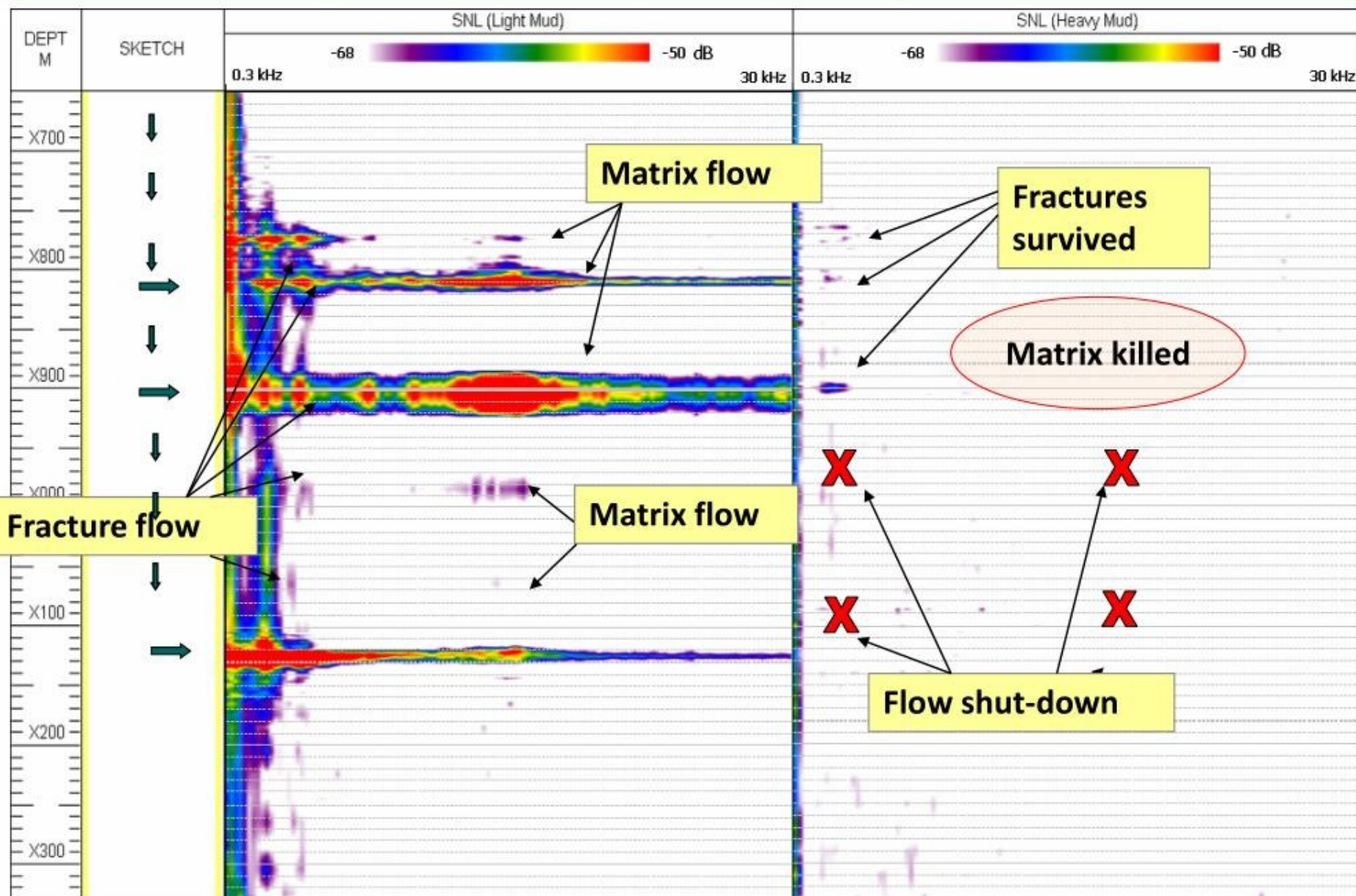


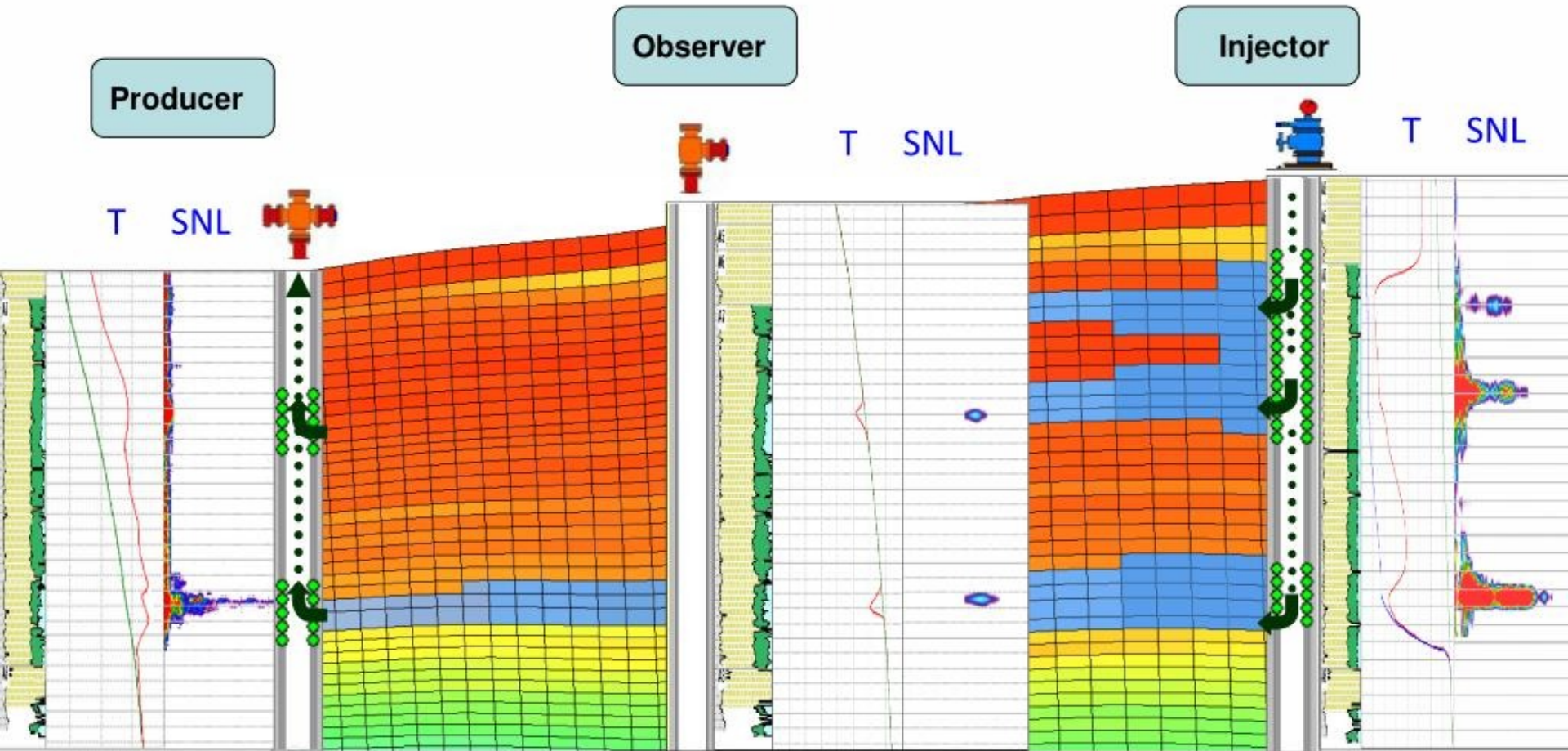
Injector. SNL + HPT + PLT. Communication





Open Hole – monitor drilling losses





The End

www.tgtoil.com

Open Hole – SNL3 vs conventional NL

